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RFO Part 12 Justification and Approval
for Restricting Competition
Simplified Procedures for Commercial Products and Services

1. Requiring Activity: US Army Corps of Engineers (USACE) Rock Island District, Brandon Road Interbasin Project.

2. Contracting Activity: US Army Corps of Engineers (USACE), Rock Island District (MVR), Rock Island, Illinois.

3. Description of Action: The USACE, MVR Contracting Office seeks approval to award a new \$3.8 million Firm Fixed Price (FFP) commercial contract for brand-name-only Lubell Labs, Inc. LL 1424 HP Underwater Acoustic Transducers in support of Brandon Road Interbasin Project (BRIP). The Contract will be fully funded utilizing Construction General funds from FY2023 appropriation.

4. Description of Supplies/Services: The Government requires 280 units of Lubell Labs, Inc. LL 1424 HP High power piezoelectric Underwater Acoustic Transducer with protective steel housing cages. This specific brand name item is essential for BRIP because its mission presents a dual challenge of extreme environmental hazards and demanding performance specifications. No other known commercial item has been proven to meet both sets of requirements.

The government's minimum needs are defined by:

a. Environmental Survivability: The transducers must operate continuously in the uniquely hostile conditions of a USACE navigation lock approach channel. This requires durability to withstand:

- Intense water turbulence from lock flushing operations.
- Constant impacts from large river debris, including submerged trees.
- Loose rock and sediment become projectiles from tow boat propellers.
- Corrosion and damage from environmental extremes and sedimentation.

b. Acoustic Performance: Simultaneously, the transducers must deliver a powerful, targeted acoustic signal. This requires the capability to:

- Ensonify a large area of water over a broad frequency range.
- Produce a specific sound pressure level (decibels) proven to deter invasive species.
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5. Authority Cited: *Authority: 41 U.S.C. 1901 as implemented in RFO 12.102(b) for acquisitions valued over the simplified acquisition threshold under the authority of simplified acquisition procedures. restricting competition (sole source) including brand name description without allowing for “equal” product.*

6. Reason for Authority Cited: In accordance with RFO 12.102 Restricting competition which states: document decisions to restrict competition, including the use of brand-name descriptions without allowing for “equal” product, (b) for acquisitions valued over the simplified acquisition threshold for commercial items, citing the statutory authority of 41 U.S.C. 1901. This determination is the result of specialized research and testing conducted by the Government's subject matter experts at the USACE Engineering Research and Development Center, Environmental Laboratory (ERDC-EL). ERDC-EL's analysis defined the unique combination of extreme durability and specific acoustic performance necessary for the BRIP. Market research efforts conducted over recent years have consistently confirmed that only the Lubell Labs, Inc. LL 1424 HP transducer possesses the specific features required to operate effectively in the mission's harsh and demanding environment.

- a. Background: For over a decade, the USACE Engineering Research and Development Center, Environmental Laboratory (ERDC-EL) has pioneered acoustic deterrent technology to combat the spread of invasive carp. This effort progressed from successful laboratory trials in flumes to a full-scale field test initiated in 2021 at Lock 19 on the Mississippi River, which was conducted in collaboration with the U.S. Geological Survey. A critical lesson from this extensive research and real-world testing is that the transducers must operate continuously (24/7/365) with near zero downtime to be effective. This imperative for extreme reliability is driven by the severe consequences of failure. Due to their placement at the bottom of a navigation channel, replacing a single failed unit is a major operation with an estimated cost of [REDACTED], requiring river closure to navigation, specialized dive teams, and crane support. To mitigate these significant operational and financial risks, the Government's minimum requirement is the specific transducer model that has already demonstrated the necessary effectiveness and durability under these proven, demanding conditions.

Justification: The ERDC-EL research and field tests described above identified several salient characteristics that are essential to meet the Government's minimum needs for performance and survivability. Market research confirms that only the Lubell brand has been proven to meet all these combined requirements. The essential characteristics are:

- a. Specific Acoustic Output: The transducer must be omnidirectional to ensure the acoustic deterrent field is gap-free, and capable of producing over 180 dB across a frequency range of 200 Hz to 8000 Hz. This specific performance profile was determined by ERDC-EL as the minimum required to effectively deter invasive carp while overcoming ambient noise from river navigation.
- b. Extreme Operational Reliability: The unit must be capable of continuous 24/7 operation while driven at 75-80 Vrms, a requirement established during long-duration

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field testing. This prevents system failure and avoids the significant cost (\$700k) and mission disruption of a replacement operation. The housing and components must be coated to, at a minimum, MIL-C-5541E standards to survive the known corrosive elements of the river environment.

- c. Essential Mission-Support Features: The transducer must feature electric wet-mate connectors. This is not a mere preference but a critical maintenance-related requirement. It enables divers to replace individual failed units underwater without needing to remove the entire array by crane, drastically reducing downtime and operational costs. The unit must also be able to operate without pressure compensation at depths between 9 and 40 feet.
 - b. Alternative: The USACE ERDC-EL conducted a comparison to alternative commercially available transducers (speakers) and found that the Lubell Lab Inc. transducers are the only model that meets the government requirement.
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